

Rung 1.5 — What even *is* a proof?

Why this rung exists: our certificate says the error is bounded "*for x in $[0, 1]$.*" That little word **for** — meaning *for every* x — is a promise you can't keep by testing. There are infinitely many x in $[0, 1]$; you'll never try them all. A **proof** is how you keep a promise about infinitely many cases at once. This rung is the one skill an EE degree skips, and everything above it leans on this.

There's no new hardware here — it's a thinking tool. It's short. Read it, then go do a dozen exercises from *Book of Proof* (Hammack, free PDF), because this is a skill, like soldering: you learn it by doing, not by reading about it.

1. The three nouns

- **Definition** — a precise agreement on what a word means. ("An integer n is *even* if $n = 2k$ for some integer k .") No fuzziness allowed; a proof is only as solid as its definitions.
- **Theorem** — a claim that is true. (Also called a lemma, proposition, corollary — same thing, different importance.)
- **Proof** — an airtight chain of reasoning from things already agreed true to the claim, with **no gaps**. "No gaps" is the whole point, and it's exactly what a computer can check (Rung 4).

2. The grammar: connectives and quantifiers

Statements are things that are true or false. You glue them together with:

- **and, or, not**
- **implies** (*if P then Q*) — the workhorse. Note: it only promises something *when P is true*. "If it rains, the ground is wet" says nothing about a dry day.

And you scope them with the two **quantifiers** — this is the part EE math never names:

- $\forall$ — "for all." *For all x in $[0, 1]$, $|error(x)| \leq 18/D + x^6$.* ← **our certificate is a $\forall$ statement.**
- $\exists$ — "there exists." *There exists a point ξ where the slope equals the average slope.* (You'll meet that exact $\exists$ on Rung 2.)

Reading a theorem is mostly figuring out its $\forall/\exists$ skeleton. Our bound, in plain words: "**for every** input x between 0 and 1, the hardware's error is at most $18/D + x^6$."

3. The three moves

Almost every proof you'll read is one of these, or a combination:

Direct proof — assume the hypothesis, march step by step to the conclusion.

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Claim: if n is even, then n^2 is even. *Proof:* n even means $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$, which is $2 \times$ an integer, so n^2 is even. ■

Proof by contradiction — assume the claim is *false*, and derive something impossible.

Claim: $\sqrt{2}$ is not a fraction. *Sketch:* suppose it were, $\sqrt{2} = a/b$ in lowest terms. Squaring gives $a^2 = 2b^2$, so a is even, so $a = 2c$, so $b^2 = 2c^2$, so b is even too — but then a/b wasn't in lowest terms. Contradiction, so no such fraction exists. ■

Induction — prove it for the first case, then prove "if it holds for k , it holds for $k+1$." Like dominoes: knock the first, and show each knocks the next, and they *all* fall — infinitely many cases from two finite arguments.

Claim: $1 + 2 + \dots + n = n(n+1)/2$. *Base:* $n=1$: $1 = 1 \cdot 2/2$. ✓ *Step:* assume it for k ; then $(1+\dots+k) + (k+1) = k(k+1)/2 + (k+1) = (k+1)(k+2)/2$. ✓ ■

4. Why this is the gate

Every single bound in this track is a **$\forall$ statement** — "for all inputs, the error is at most ...". You now know what it *means* to prove one, and which of the three moves a proof is using. That's enough to start reading real proofs slowly.

And here's the punchline that connects to Rung 4: a proof assistant like **Lean** is just a very strict, very patient computer that checks each step of these moves and refuses to let you skip one. "Machine-checked" = a computer confirmed there are no gaps. Everything you're learning here is what you'll be telling the machine.

5. Homework (do this one, don't skip it)

- *Book of Proof* (Hammack), Chapters 1–4: sets, logic, quantifiers, direct proof. Do the exercises with a pen. If you want a second voice, *How to Prove It* (Velleman).
- Goal: be able to read a $\forall/\exists$ statement and say out loud what it's promising, and write a three-line direct proof without freezing.

You don't need to be fast or fluent — just un-scared. Once "for all x , ..." reads like a sentence instead of a wall, you're ready.

→ **Next:** Rung 2 — *Where the x^6 comes from* — your first real $\forall$ proof about actual hardware, using exactly this grammar.

Exercises & checkpoint

1. Rewrite our certificate — "the hardware error is at most $18/D + x^6$ for every x between 0 and 1" — using the symbols $\forall$, $\exists$, and $\rightarrow$.

2. Prove directly: if n is odd, then n^2 is odd. (Start: $n = 2k+1$.)

3. From *Book of Proof*, do five exercises from the quantifiers chapter — with a pen.

Checkpoint: you can read a $\forall / \exists$ statement aloud as a plain sentence, and write a three-line direct proof without freezing.